Varun Ravi Varma

AI Research Scientist & AI / Research Engineer

Grimstad, Norway | varunravivarma@gmail.com | v3rm1.github.io
GitHub: v3rm1 | ORCID: 0000-0003-1719-2478 | LinkedIn: varun-ravi-varma

Availability & Languages

Availability: Research programmer and applied scientist roles in Germany, Austria or Switzerland (DACH) from early 2028
Languages: English (C2 – Proficient), German (A2–B1, currently taking courses at Heidelberg University)

Professional Summary

AI Research Scientist and Software Engineer with over 5 years of experience bridging theoretical machine learning breakthroughs and production-grade software architectures. Specialised in GPU-accelerated simulation platforms, multi-agent reinforcement learning (MARL), LLM/agentic workflows, and semantic knowledge graph pipelines. Demonstrated track record of transforming complex mathematical formalisms into high-throughput, reproducible research software across distributed GPU clusters (HPC) and enterprise cloud environments. Experienced in designing robust data pipelines, establishing CI/CD automation, and deploying resilient machine learning systems in healthcare, finance, and industrial domains.

Technical Skills

Programming Languages: **Python**, **C++**, **Rust**, **SQL**, **CUDA**, **Bash**, **PowerShell**
AI Frameworks & Modelling: **PyTorch**, **JAX**, **Multi-Agent Systems (MARL)**, **LLMs & Agentic Architectures**, **Graph Neural Networks**, **Scientific ML**, **Non-Linear Dynamical Systems**
Systems, Cloud & MLOps: **Docker**, **Kubernetes**, **Git-based CI/CD**, **Linux Server Administration**, **Distributed GPU Clusters (Slurm/HPC)**, **Azure**, **Databricks**, **Model Serving (REST/gRPC)**

Professional Experience

Doctoral Research Fellow (AI & Complex Systems) 09/2023 – Present
University of Agder *Grimstad, Norway*

- Research and implement multi-agent reinforcement learning (MARL) algorithms for coordinated, long-horizon decision-making in complex multi-agent systems.
- Architected a high-throughput, GPU-vectorised simulation engine in JAX (`jit`, `vmap`) that steps thousands of scenarios in parallel on a single GPU.
- Designed an automated validation suite to check simulated dynamics, parameter sensitivity, and policy behaviour, with reproducible multi-seed experiment sweeps on multi-GPU servers.
- Mentored student researchers in modular software design, version control workflows, unit testing, and reproducible code standards for scientific machine learning.
- Directed advanced scientific writing and technical presentation workshops across university departments.

Scientific Programmer 06/2022 – 08/2023
Sano Centre for Computational Medicine *Krakow, Poland*

- Architected analytical data pipelines and knowledge graph reasoning frameworks for clinical decision support and complex biomedical artifact analysis.
- Designed relational schemas and semantic dependency mapping algorithms to structure heterogeneous medical data into queryable knowledge graphs.
- Developed and containerised reproducible computational tools and model evaluation harnesses using Docker to ensure seamless cross-platform execution.
- Implemented Git-based CI/CD workflows to automate regression testing, model training pipelines, and data integrity verification in regulated environments.
- Collaborated with clinicians and domain scientists to translate rigorous clinical requirements into production-grade software architectures.

Teaching Assistant 08/2025 – 01/2026
University of Agder *Grimstad, Norway*

- Research Methods for AI and IoT: assisted with lectures, tutorials and exam evaluations.
- Computer Vision: assisted with lectures and labs, mentored student projects and evaluated technical reports.

Graduate Teaching Assistant 02/2020 – 08/2020
University of Groningen *Groningen, Netherlands*

- Deep Learning: assisted with labs, mentored students and evaluated technical practical reports.
- Logical Aspects of Multi-Agent Systems: facilitated tutorials, graded assignments and reviewed course projects.

Consultant – Data Analytics & Machine Learning 06/2016 – 11/2018
Neudesic Global Services *Bangalore, India*

- Architected and deployed enterprise-scale data analytics pipelines, deep learning models, and high-capacity data warehousing solutions for Fortune 500 clients.
- Developed proof-of-concept machine learning solutions across energy, pharmaceutical, and financial sectors.
- Managed migration and integration pipelines for structured and unstructured data across hybrid cloud and on-premise high-availability infrastructures.
- Founded and led an internal technical upskilling initiative, directing weekly seminars on statistics, machine learning, and scalable systems.
- Automated maintenance procedures and system monitoring via scripting to ensure 99.9% operational uptime.

Key Research & Engineering Projects

GPU-Accelerated Multi-Agent Simulation | *JAX, Python* | *Ongoing, details after publication*

- Building fully vectorised, JIT-compiled simulation environments in JAX so that thousands of scenarios step in parallel on a single GPU.
- Validation suite to check that the simulated dynamics behave plausibly.

Hierarchical Multi-Agent Reinforcement Learning | *JAX, PyTorch, Python* | *Ongoing, details after publication*

- Designing structured, hierarchical MARL architectures in which agents coordinate in sub-teams under partial observability and long horizons.
- Compared against established flat MARL baselines across many random seeds with statistically sound aggregate metrics.

Clinical Knowledge Graph & Semantic Reasoning Engine | *Python, Neo4j, Docker, CI/CD*

- Built end-to-end knowledge extraction and semantic reasoning graph pipelines unifying structured electronic health records and unstructured medical literature.
- Containerised pipeline components with Docker and integrated automated validation testing into continuous integration workflows.

Education

Ph.D. in Artificial Intelligence 2023 – Present
University of Agder *Grimstad, Norway*
Focus: Multi-Agent Reinforcement Learning, GPU-Accelerated Simulations, and Complex Systems

M.Sc. in Artificial Intelligence 2019 – 2021
University of Groningen *Groningen, Netherlands*

B.Tech. in Mechanical Engineering 2012 – 2016
Visveswaraya Technological University *Belagavi, India*